

Audio Anomaly Detection on Water Pumps

Learning a ConvAE Latent Space for Unsupervised Fault Detection

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What is Acoustic Anomaly Detection?

Condition Monitoring

Detect machine faults early, directly from the sound they emit.

Unsupervised / One-Class

Real deployments only have healthy audio; faults are rare and unlabeled.

Represent → Detect

First learn a feature space, then flag samples that fall outside it.

Water-Pump Audio Dataset

4

machine IDs (00 / 02 / 04 / 06)

16 kHz

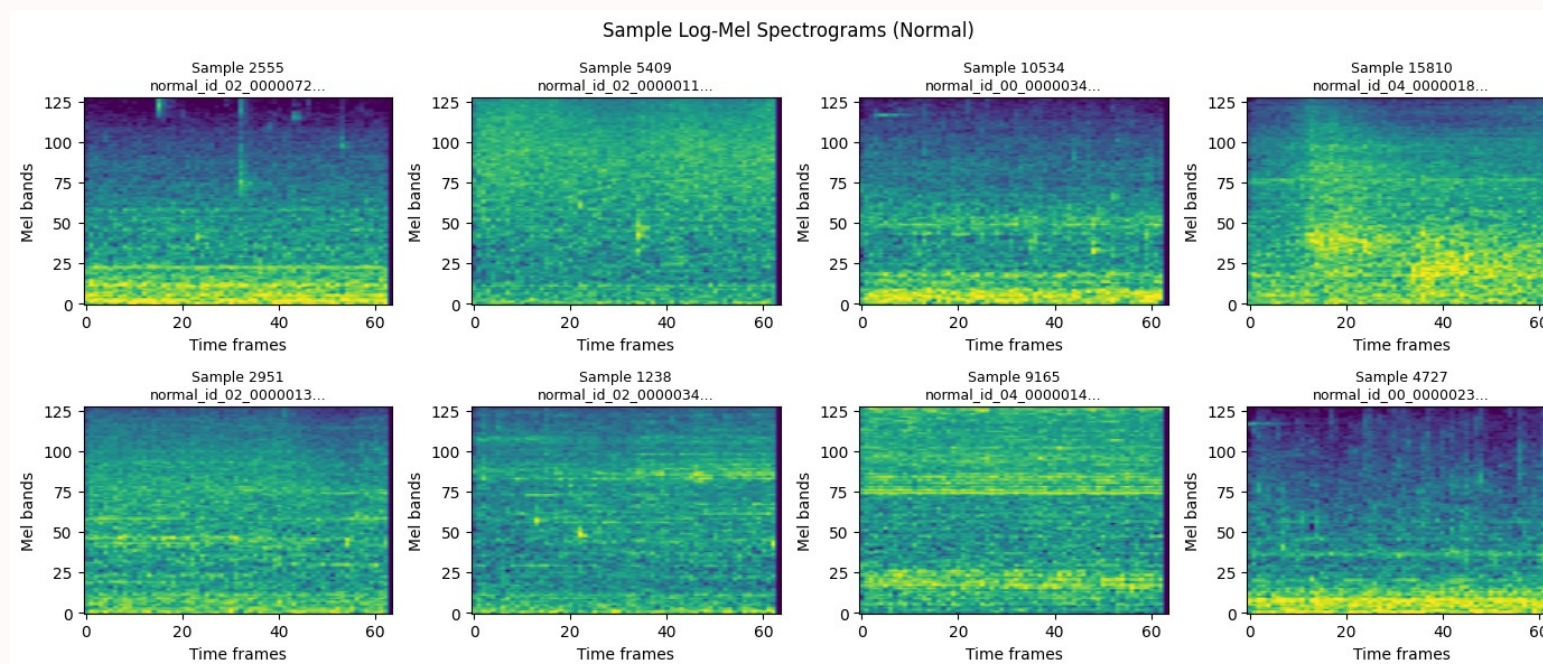
mono audio

9

windows per 10 s clip

200 / 320

normal / anomaly test files



ID 06 is never seen by the AE or the detectors — it appears only in tuning and test data. File-level splits, seed 42.

Data & Preprocessing

From Waveform to Spectrogram

1

Load audio — mono,
16 kHz

2

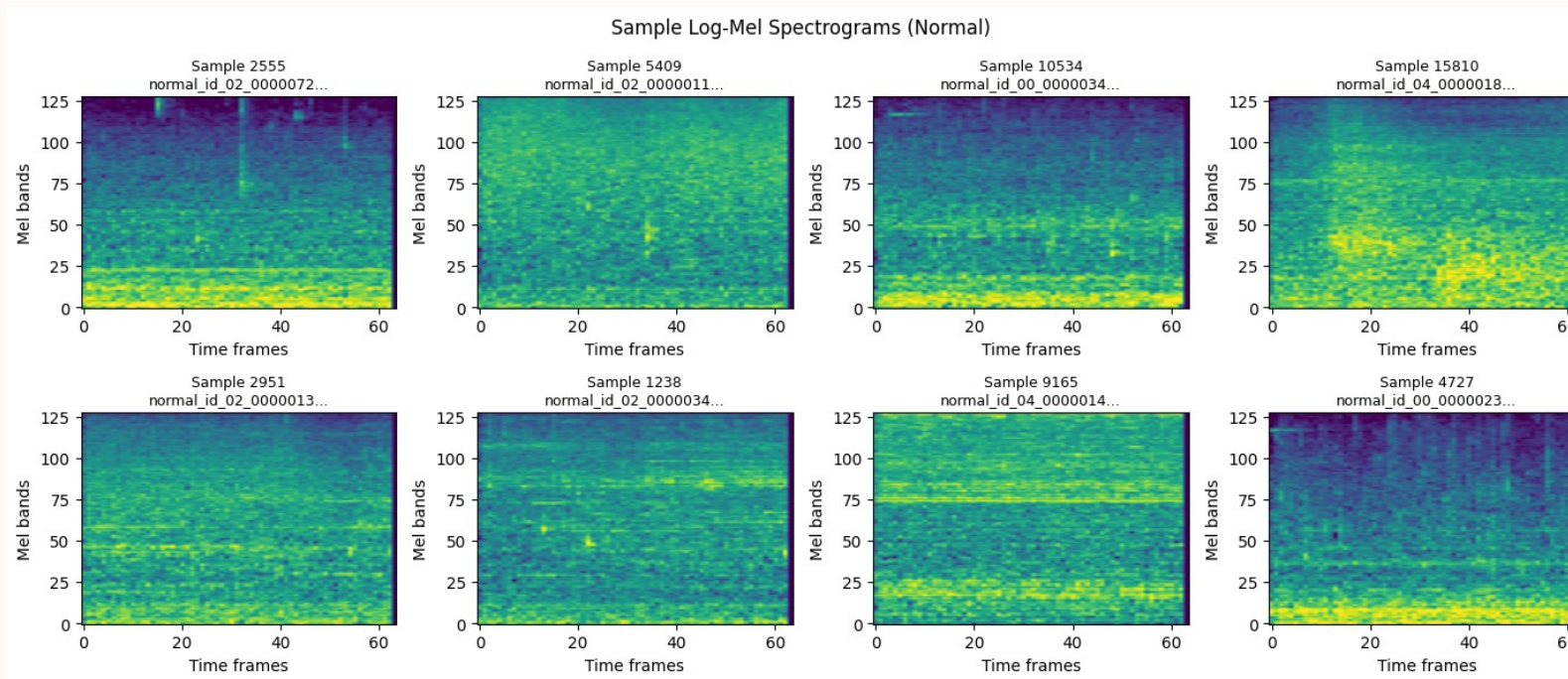
Slice — 2 s windows,
1 s hop (9 / clip)

3

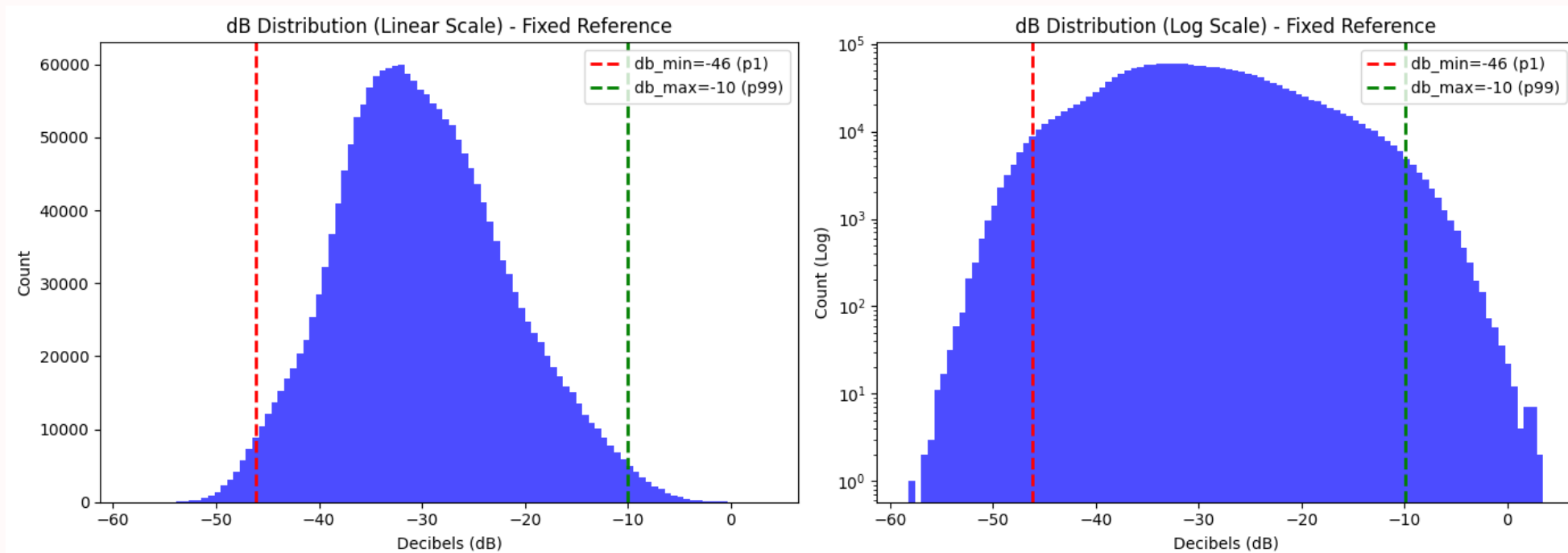
Log-mel
spectrogram — 128
× 64

4

Normalize to [0, 1]



A Normalization Choice That Matters



**Fixed reference
preserves loudness**

- Convert to dB with $\text{ref} = 1.0$, not per-image max.
- A per-image max would erase absolute intensity across files.
- Clip to 1st / 99th percentile ($\approx -46 / -10$ dB), then map to $[0, 1]$.

The ConvAE Representation

MODEL

Convolutional Autoencoder



Encoder (spectrogram $\rightarrow$ latent)

Decoder (latent $\rightarrow$ reconstruction)

8192 $\rightarrow$ 64

input values $\rightarrow$ latent

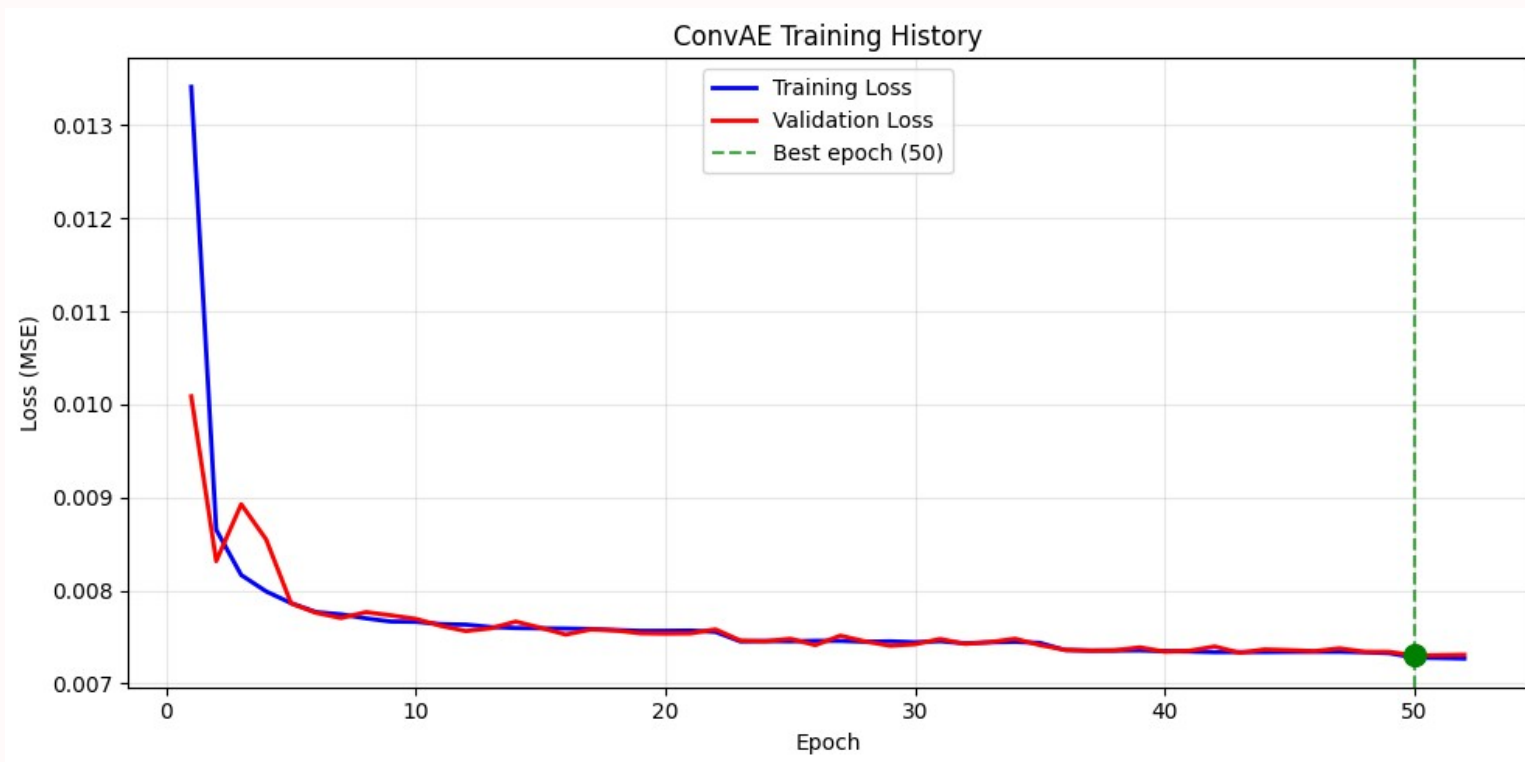
128 $\times$

compression ratio

NORMAL only

training data

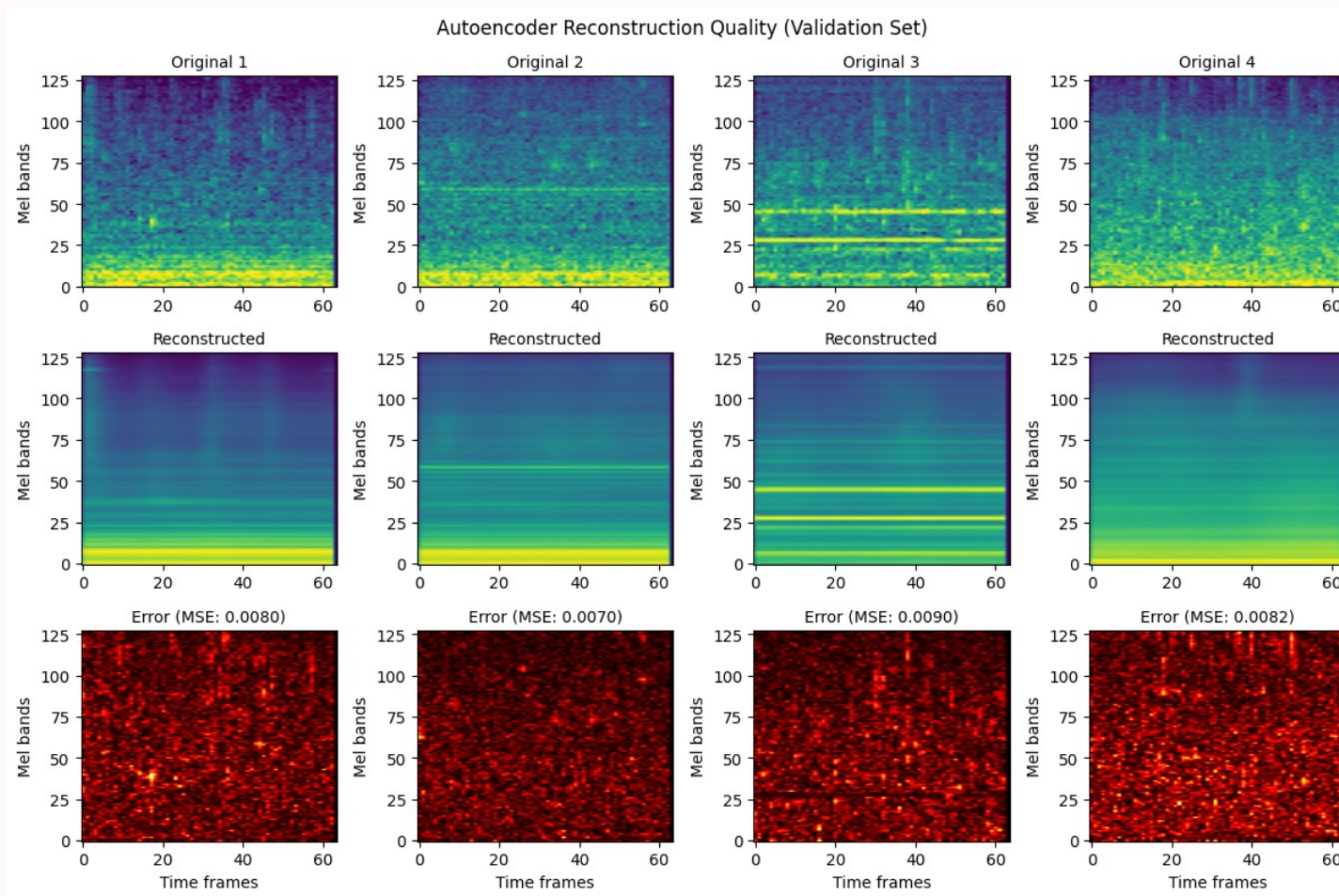
Training Converges Cleanly



Clean convergence

- 52 epochs, best validation MSE ≈ 0.0073 .
- Early stopping + reduce-on-plateau scheduler.
- Train and validation track closely — no training failure.

Reconstruction Quality



The AE rebuilds spectrograms well

- Faithful reconstructions on normal windows, as expected.
- But it also reconstructs many anomalies acceptably...
- ...foreshadowing why plain reconstruction error is a weak score.

Detecting Anomalies

Two Families of Detectors

All operate on the frozen ConvAE (latent vector or reconstruction error).

Unsupervised

- Reconstruction Error (per-window MSE)
- Isolation Forest (latent)
- Local Outlier Factor — LOF (latent)
- One-Class SVM (latent)

Supervised

- SVM-RBF (latent)
- XGBoost (latent)
- SVM on MFCC statistics

Detectors on the Latent Space

One-Class SVM

Learns a single rigid boundary around normal data.

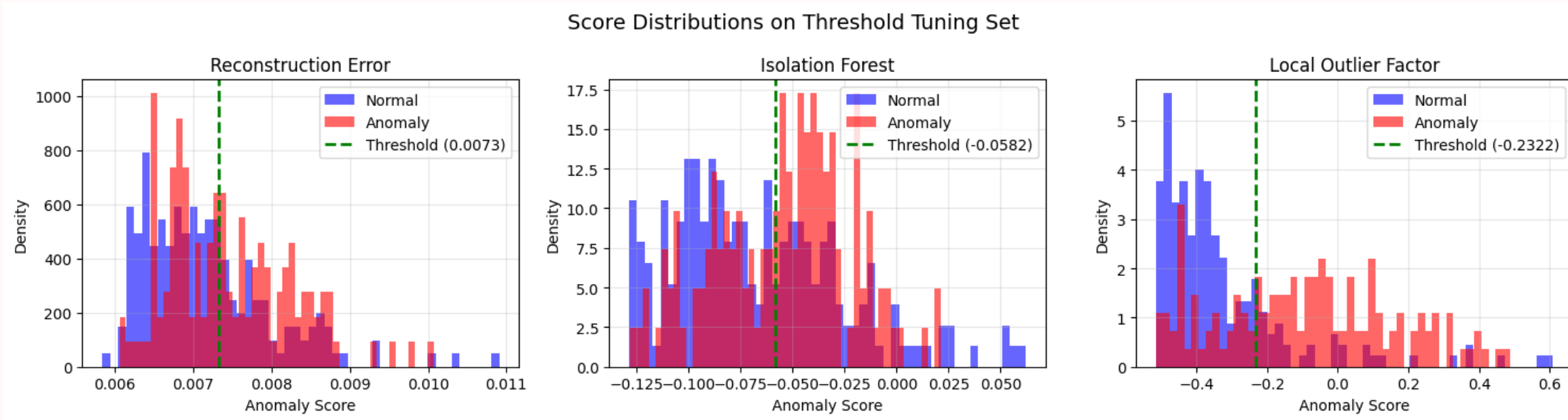
Local Outlier Factor

Density-based: flags points in low-density regions.

Isolation Forest

Isolates outliers with random splits in feature space.

Thresholding & Evaluation



- Window scores → file scores by mean aggregation.
- Threshold tuned only on the tuning split (balanced accuracy).
- Final test untouched; report AUC-ROC / PR (threshold-free).

Results & Analysis

RESULTS

Winner: ConvAE + LOF

73.3%

Balanced Accuracy

held-out file-level test

0.788

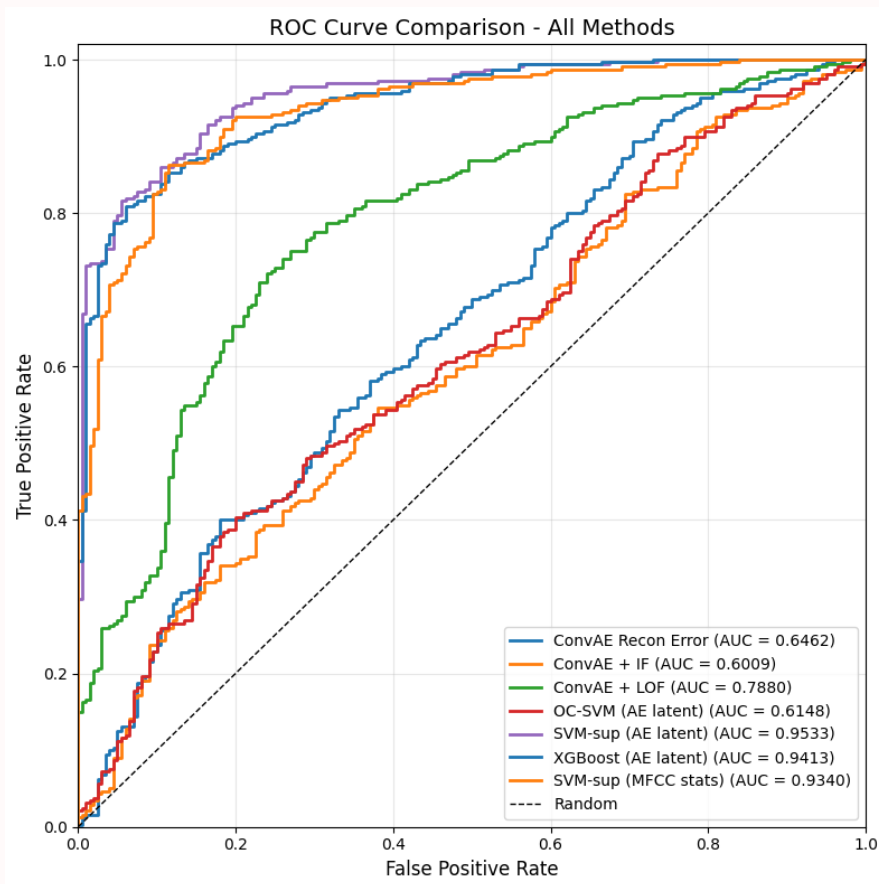
AUC-ROC

threshold-independent

Best assignment-aligned unsupervised method.

RESULTS

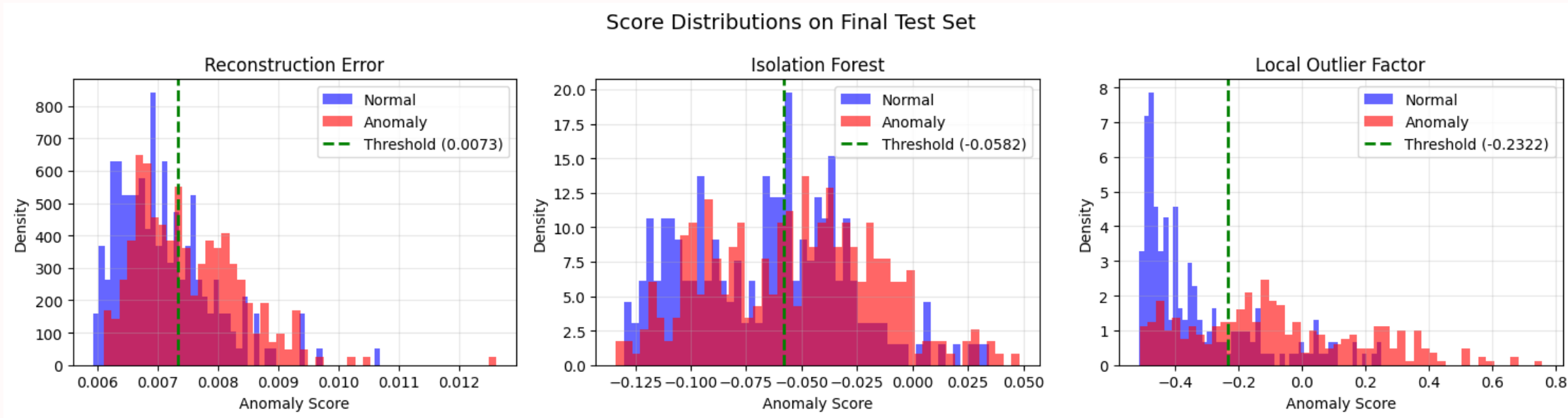
All Methods Compared



Method	Bal Acc	AUC
ConvAE Recon Error	0.595	0.646
ConvAE + IF	0.560	0.601
ConvAE + LOF	0.733	0.788
One-Class SVM	0.545	0.615
PANNs + LOF	0.702	0.761
SVM-sup (latent)	0.827	0.953
XGBoost (latent)	0.827	0.941
SVM-sup (MFCC)	0.854	0.934

Grey rows use anomaly labels (diagnostic baselines).

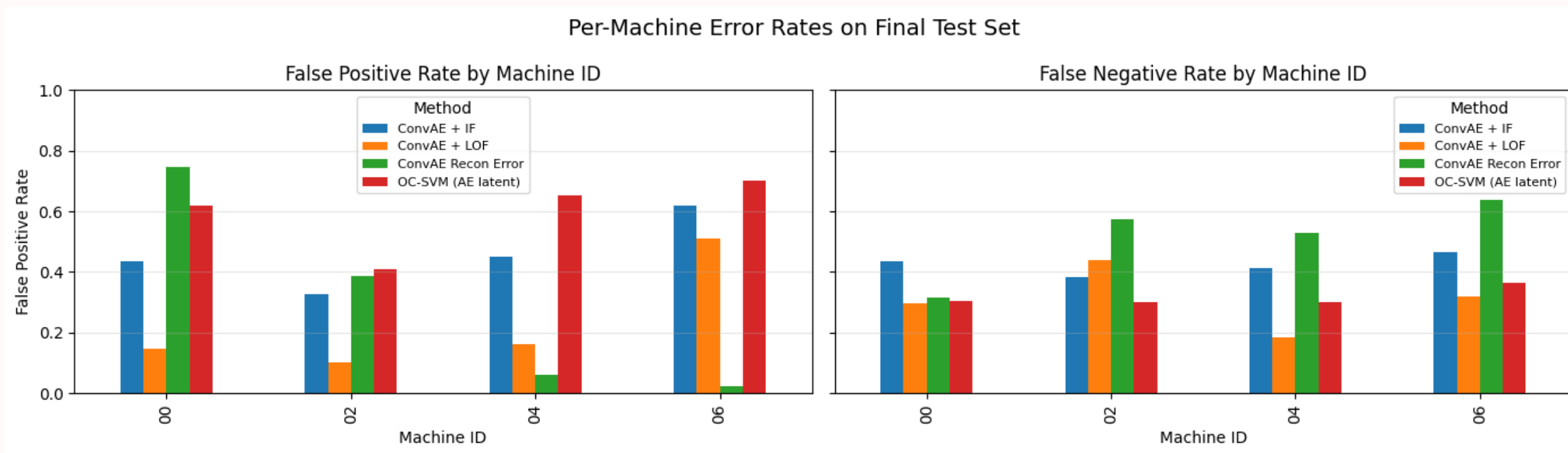
Why Reconstruction Error Is Weak



Overlap vs separation

- Reconstruction-error and IF scores overlap heavily.
- LOF separates normal and anomaly files best.
- Average pixelwise MSE misses local, textural faults.

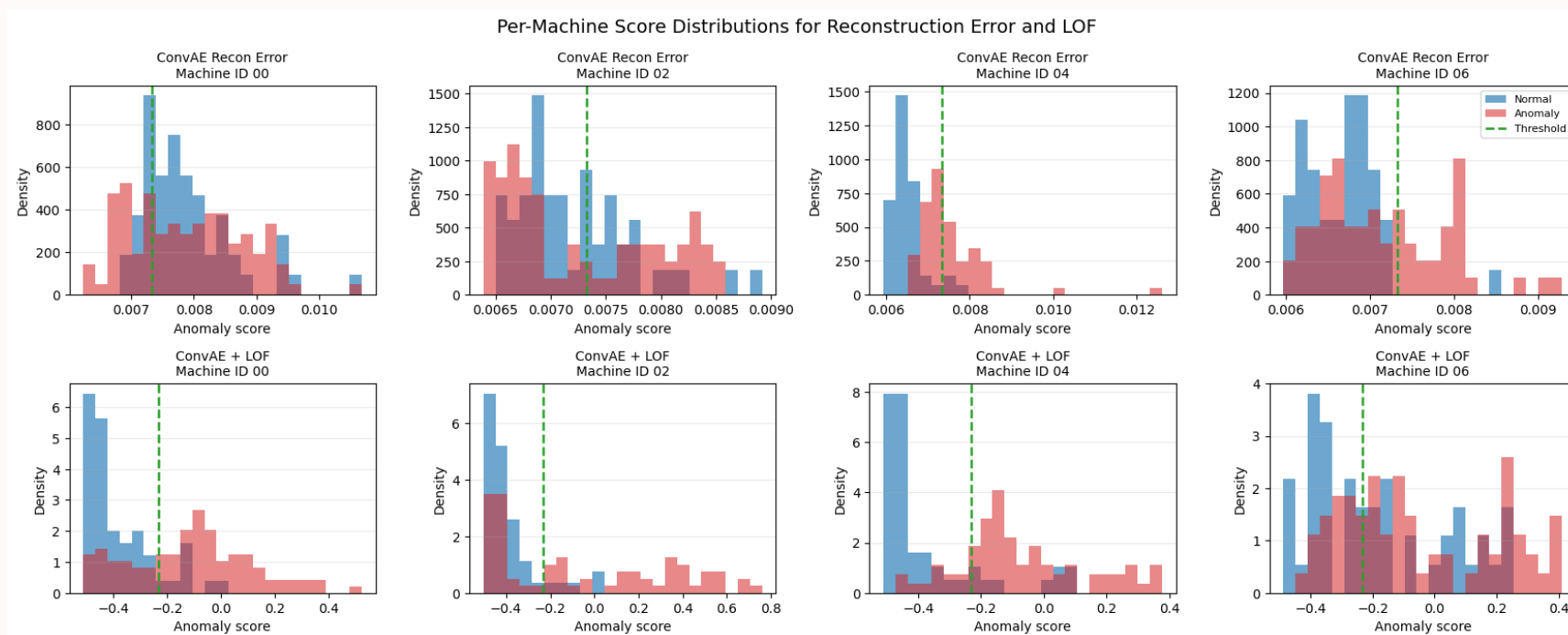
The ID 06 Problem



One unseen machine drives false alarms

- ID 06 LOF false-positive rate: 51.1%.
- Other IDs: 14.5% / 10.2% / 16.3%.
- The model confuses a new normal machine with an anomaly.

Per-Machine Score Distributions



- ID 06 normal LOF scores straddle the threshold — the 51% FPR, visually.
- For recon error, ID 06 normals sit below threshold (lowest FPR).
- The ID 06 problem lives in the latent-density model, not reconstruction.

Ablation — Add ID 06 Normals (Protocol 2)

51% → 40%

ID 06 false-positive rate

lower is better

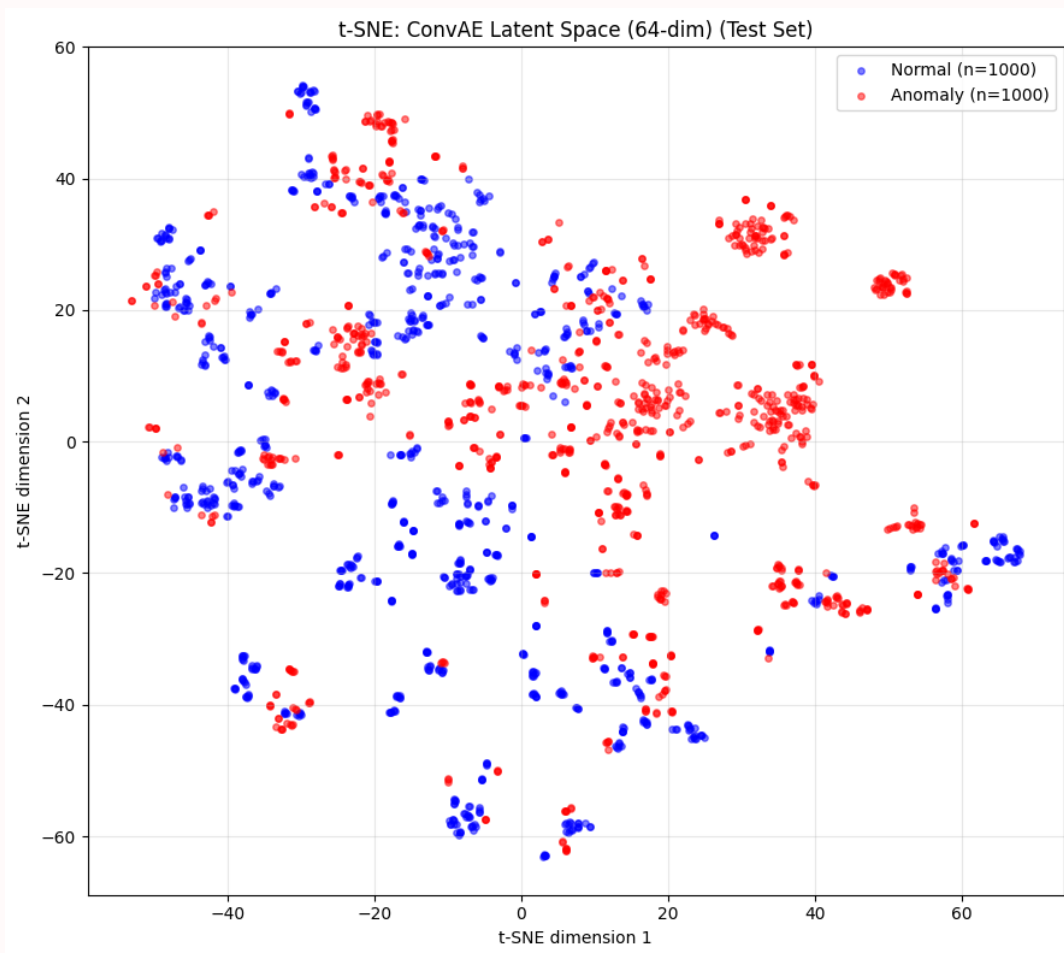
0.79 → 0.81

LOF AUC-ROC

slight overall gain

ID 06 normals taken from the tuning split — test set unchanged. FNR rises 31.9% → 34.8%: overlap remains, so unseen-machine variation is only a partial cause.

The Latent Space Is Only Partly Separable



Qualitative view

- t-SNE of ConvAE latents on the test set.
- Normal and anomaly points partially overlap.
- Consistent with the moderate unsupervised scores.

A Diagnostics Recap

✓ Normalization

Fixed dB ref = 1.0, clipped to the training 1st / 99th percentiles (−46 / −10 dB) — absolute loudness preserved, verified on the training distribution.

✓ Tuning & aggregation

4 aggregation strategies × 3 tuning metrics swept — mean is the most stable. Thresholds tuned on a separate split, no leakage.

✓ MFCC sanity baseline

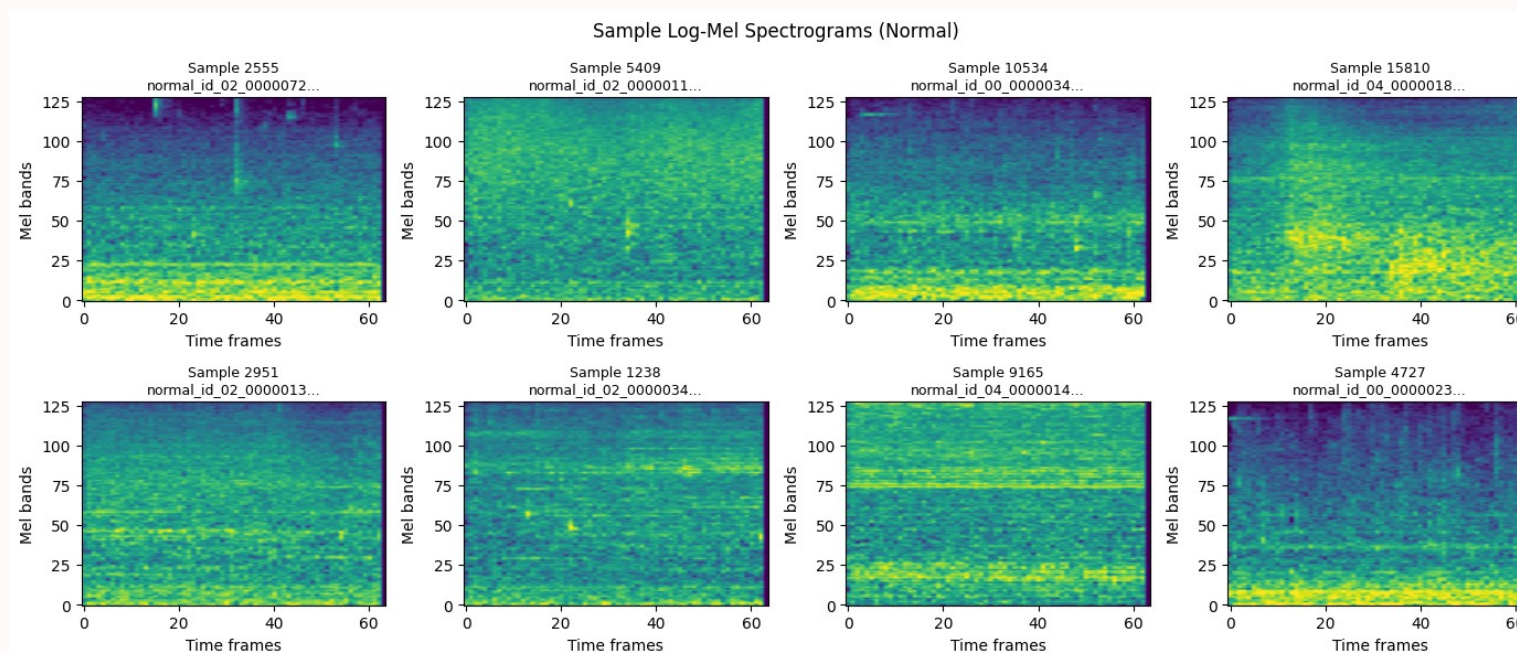
With anomaly labels, an SVM on hand-crafted MFCC statistics reaches 0.854 BA (AUC 0.93) — the signal exists; the unsupervised gap is not missing information.

✓ Unseen-machine ablation

Adding ID 06 normals to training helps only partially (FPR 51% → 40%, FNR rises) — genuine overlap remains.

→ **The signal exists — but without labels, subtle faults overlap with normal machine variation.**

Conclusion



Achieved

- ConvAE latent space + LOF: 0.733 BA / 0.788 AUC, fully unsupervised.
- Reconstruction error alone is insufficient for subtle pump faults.
- Diagnosed the unseen-machine (ID 06) false-positive effect.

Thank you.

Questions?